

ject, Indian Air Force has developed Air Force Network (AFNet) at a cost of 10.77 billion rupees in collaboration with HCL Infosystems Ltd and Bharat Sanchar Nigam Ltd (BSNL). This gives hope for better involvement of domestic firms in the future.

Capital investment. Establishing a firm in strategic electronics industry requires huge capital expenditure during initial years, which is a hurdle for start-ups, small and medium companies. Time delay in approval processes increases the cost further.

Technology. The inherent ecosystem of India doesn't support the fast adoption of cutting-edge technology. Hence, we have to depend on the expensive technologies and machines from abroad.

Slow tender process. The closure of tender cycle in Indian defence contracts sometimes takes too long—about 18 to 24 months. In order to encourage private players' active participation in defence opportunities, this should be kept to within a reasonable 'start-end' period.

Testing. Lack of good test infrastructure for strategic electronics is an issue. Quality labs are limited, which makes them always occupied. The government should set up a number of testing centres.

Tax issues. Duty-free imports and high taxes on indigenous products make the vendors prefer foreign equipment over Indian brands. There is a need for elimination of indirect taxes in defence field.

Ambiguity of policies. Though the government is coming up with policies and tenders to promote A&D industry, many of these do not have a clear framework. For example, in the case of dual-use equipment, there arise issues like whether or not they require an industrial licence. The recently modified industrial licensing policy is expected to solve this matter.

Recommendations of industry

Despite the positive approach of government towards developing A&D

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industry, we still face many limitations and roadblocks. The following are some of the recommendations of the industry.

Raising FDI. The previously 26% foreign direct investment (FDI) cap was a major turn-down for foreign players to enter this industry. This has been increased to 49% in the latest budget approved on 18th July.

Ecosystem development. The government has to aid the creation of infrastructure and adoption of latest technologies required for manufacturing high-quality components that would cater to domestic as well as foreign markets. They should encourage and provide opportunities for Indian design and manufacturing houses to compete with foreign players. This can be achieved by providing government incentives or preferences, such as waivers in taxes and duties.

Reducing import. Though immediate reduction in imports is quite difficult, channelised imports are important to boost self-reliance in strategic electronics. A clear roadmap should be created for import reduction by the Ministry of Defence, which should be published and audited by the Comptroller and Auditor General (CAG) of India, with aggressive targets for the information and communication technology in education (ICTE) sector.

Investments in components and composite. It is important to invest in and promote components and composites in India. No 'end-use' restriction should be imposed so as to ensure more participation of Indian firms in strategic electronics.

Speed up decision-making: It often takes up to five years to get an

industrial licence. The huge delay in granting and renewing of licence, the complex procedures involved and corruption negatively affects the growth of industry, as a lot of extra costs are incurred due to the time overrun. ELCINA association had requested the government to ensure that the decision-making process be made faster, as MNCs are unable to wait for too long.

The future

According to Gautam Awasthi of Agilent, "Globally, aerospace and defence markets are in a growth phase primarily due to investments by countries like India, Russia, China and South East Asia. USA and Europe markets have been consolidating to meet their strategic defence guidance."

Aerospace and defence market is expected to reach US\$ 248.2 billion globally by 2018 as per *Aerospace-Defence Electronics-Global Trends, Estimates and Forecasts, 2011 to 2018*. Geographical analysis shows that the highest CAGR of 14.2% is anticipated from Asia-Pacific region during the analysis period, 2011-2018.

We foresee that electronic warfare will continue its growth and more investment would be made in building a robust communication backbone. Satellite technologies will also trigger growth for the sector. RF surveillance is a growing trend in keeping the country safe. Radars will continue to evolve and within that there will be subsidiary trends like transmitter-receiver modules. ●

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